

hw 1

Ian Chi

January 13, 2026

1. (a) the sum over all x and y should add up to 1.

$$\sum_{x \in \{0,1,2\}} \sum_{y \in \{0,1\}} c(x+y) = 1 \quad (1)$$

this implies $c = \left(\sum_{x \in \{0,1,2\}} \sum_{y \in \{0,1\}} (x+y) \right)^{-1} c = 1/9$

(b)

$$P(x) = \sum_y c(x+y) = c(x+1) \quad (2)$$

$$P(y) = \sum_x c(x+y) = c(y+3) \quad (3)$$

(c)

$$P(x|y=1) = \sum_x c_2(x+1) \quad (4)$$

where $c_2 = 1/6$

- (d) no. the distribution for x changes based on whether y is 0 or 1.